

Uncertainty-routed skin-lesion triage, and an audit for the ink-marking shortcut

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Summary

AMSDDS is a two-stage system for 7-class dermatology triage on HAM10000 and PAD-UFES-20. A fast MobileNetV3 handles confident cases; a calibrated confidence + entropy gate sends uncertain ones to a fine-tuned PanDerm ViT-B/16. It started as a five-person hackathon project (Smart Horizon 2026, top-5 finalist). After the hackathon I kept working on it alone: I rebuilt the escalation model, added a smartphone-photo head, and, after reading Janizek et al.'s Visual Concept Ranking paper (MLHC 2026), audited the model for the biopsy-ink shortcut that paper describes.

Headline result	Number
Cascade accuracy vs. running the transformer on every image	86.9% vs 86.4%, at ~30% of its compute
Gate catches: malignant lesions the fast path called benign	54 of 75
Calibration (expected calibration error)	0.025
Escalation model rebuild: macro F1	0.736 → 0.788
Dermoscopy → smartphone transfer: cross-domain macro F1	0.550 → 0.704
Ink audit: benign lesions whose P(malignant) rose, purple vs grey dots	83% vs 49%
Ink audit: pushed past the deployed sens-95% threshold, purple vs grey	7.1% vs 3.1%

Who did what

Phase	Work	Owner
Hackathon (5-person team)	Cascade design, Layer 1 MobileNetV3, Flask/Docker serving, demo	Team
Hackathon	Dataset acquisition, cleaning and lesion-grouped splits; shared preprocessing; Flask decision engine; OOD (Mahalanobis) gate; first Layer 2 (RT-DETR encoder + head)	Me
After the hackathon	PanDerm ViT-B/16 rebuild of Layer 2; PAD-UFES-20 smartphone head; ink-marking audit (this document)	Me, solo

System

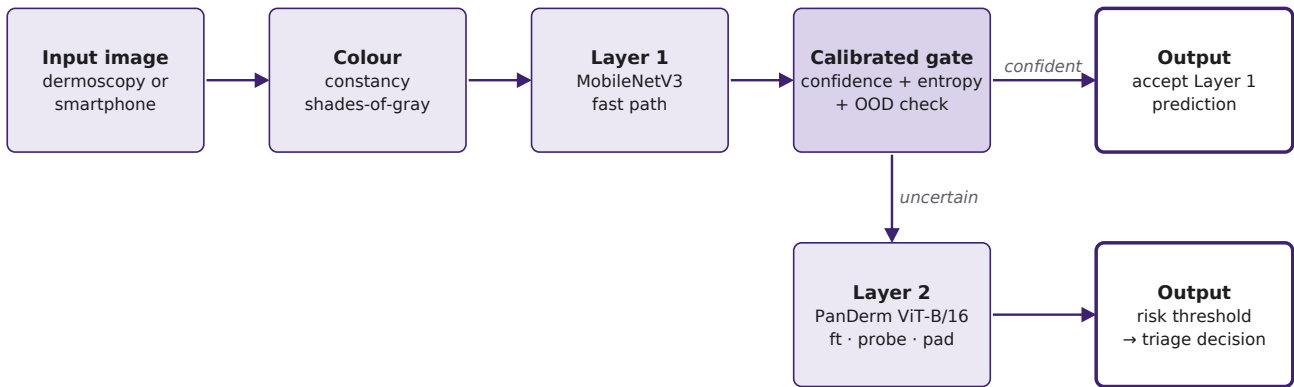


Figure 1. The cascade. Images that Layer 1 is confident about exit early; the rest go to Layer 2, whose calibrated $P(\text{malignant})$ is compared against a risk threshold chosen for a target sensitivity. Layer 2 has three selectable heads: *ft* (fully fine-tuned), *probe* (frozen features) and *pad* (smartphone photos).

Data and splits

HAM10000 (10,015 dermoscopy images) is split by **lesion**, so no lesion appears in two splits; a leakage assert runs before training. Train / val / test = 7,116 / 1,458 / 1,441. The test set is heavily imbalanced: 974 nevi versus 9 dermatofibromas. PAD-UFES-20 (smartphone clinical photos) is split by **patient** with a 6→7 class mapping. One judgment call is flagged openly: PAD's invasive SCC is mapped to HAM's *akiec* (same keratinocyte carcinoma family, different stage), and PAD has no df or vasc images.

Why Layer 2 was rebuilt

The first escalation model, a COCO-pretrained RT-DETR encoder with a classification head, beat the fast path by only 1.2 points. My read was that a detection backbone learns object-versus-background separation, while lesion classification depends on fine texture. I replaced it with PanDerm_Base (a dermatology foundation model, BEiT ViT-B/16 at 224 px) and fine-tuned it fully on a single T4.

Setting	Value
Preprocessing	Shades-of-gray colour constancy ($p = 6$), resize short side 255, centre crop 224, ImageNet normalisation
Optimisation	AdamW, lr 5e-4, weight decay 0.05, layer-wise LR decay 0.65, 10 warm-up epochs, batch 128, drop-path 0.2
Imbalance and regularisation	Class-balanced sampler, mixup 0.8, cutmix 1.0
Schedule	Planned 50 epochs, stopped at 40; best validation macro F1 at epoch 37
Calibration and inference	Temperature scaling fitted on validation ($T = 0.827$), 4-view test-time augmentation
Test result	Accuracy 0.851; macro F1 0.788; the two rarest classes (89 training images each) reach 0.842 and 0.927 F1

Before fine-tuning I locked in a frozen-probe result, so there was a defensible fallback if fine-tuning failed. Cached frozen features cut experiment turnaround from about 50 minutes to seconds, which made a 24-configuration sweep practical; configurations were selected on validation and the test set was reported once.

Smartphone photos

Dermoscopy and smartphone photos look very different, so the HAM-trained heads cannot be trusted on phone images. I trained a separate head on PAD-UFES-20 and compared two encoders: pretrained PanDerm and the HAM-fine-tuned one. The HAM-fine-tuned encoder transferred better, which suggests dermoscopy fine-tuning helps clinical-photo transfer rather than narrowing the features. That motivated a shared-backbone, per-domain-head design. Cross-domain macro F1 went from 0.550 to 0.704.

The ink-marking audit

Motivation. Janizek, Xu, Lateef and Daneshjou (MLHC 2026) used Visual Concept Ranking to show that large multimodal models use blue/purple biopsy ink as a malignancy shortcut, and that this reliance changes with skin type and prompt. My model is a fine-tuned vision transformer rather than an LMM, so VCR does not apply directly, but the paper’s validation step, adding the suspected feature by hand and measuring the change, does. I ran that intervention.

Method

Step	Detail
1. Reproduce first	Loaded the exact checkpoint and ran it through my own preprocessing and TTA code. Test accuracy reproduced the saved 0.8508 exactly before any experiment ran. (An earlier attempt with re-implemented preprocessing was 6 points off; I discarded it.)
2. Select images	All 1,064 test lesions that are benign <i>and</i> correctly called benign by the model.
3. Three conditions	Clean: JPEG re-save only (q95). Purple: four dots, RGB (60, 34, 112), the colour used in the paper. Grey: identical dots in identical positions, RGB (128, 128, 128). The grey control separates “ink” from “any blob”.
4. Placement	Dots drawn on the raw photo <i>before</i> colour constancy, as real ink would be. Radius 4% of the visible crop, placed in the left and right bands of the region that survives the centre crop, away from the (usually central) lesion.
5. Measure	P(malignant) = sum of calibrated mel + bcc + akiec probabilities. Paired change per image, bootstrap 95% CI, Wilcoxon signed-rank test, and crossings of the deployed 95%-sensitivity risk threshold.

Results

Condition	Mean P(mal)	Mean Δ (95% CI)	P(mal) rose	Crossed sens-95% thr.	Argmax flip	Wilcoxon p vs clean
Clean (JPEG)	0.053	—	—	—	0.7%	—
Purple ink	0.064	+0.011 [0.008, 0.015]	83.0%	7.1%	1.2%	< 1e-10
Grey control	0.061	+0.008 [0.005, 0.011]	48.7%	3.1%	1.5%	0.006

Purple versus grey, paired on the same images: mean difference +0.0035, Wilcoxon p = 3.8×10^{-40} .

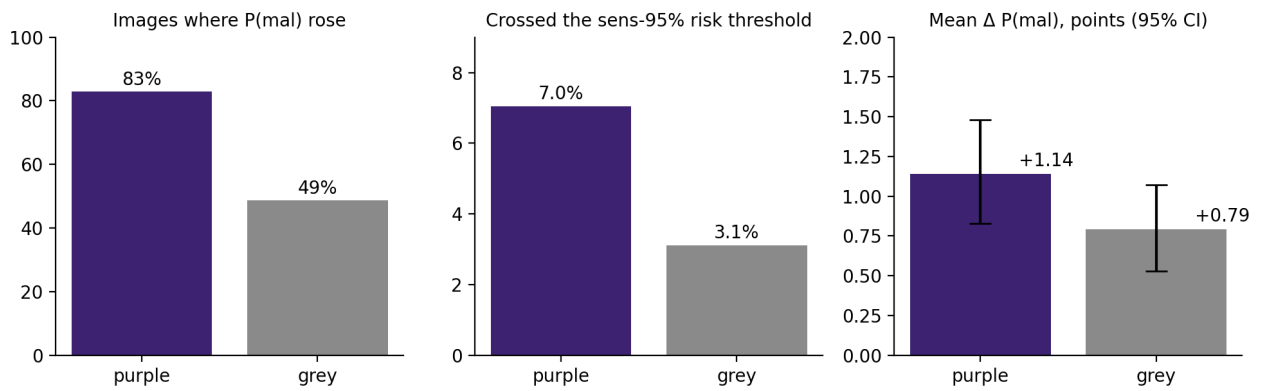


Figure 2. Purple ink raises the malignancy estimate far more consistently than grey dots in the same spots, and pushes over twice as many benign lesions past the threshold used for triage.

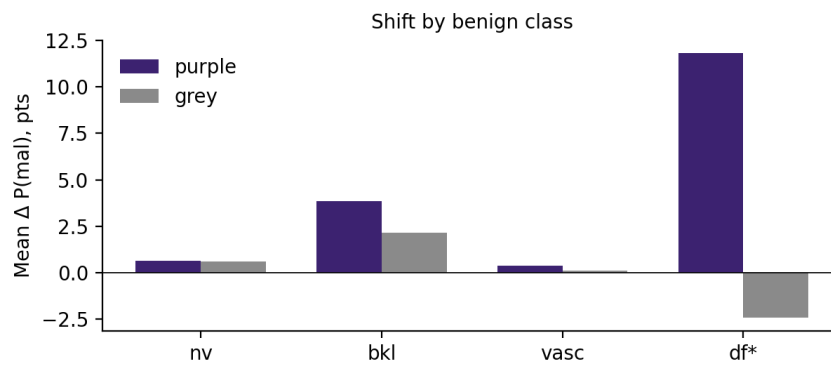


Figure 3. Mean change by benign class. Nevi barely move under either condition; keratoses (bkl) react more to purple than grey. *Dermatofibroma has at most 9 test images, so its large value is not reliable and I do not draw conclusions from it.

Examples

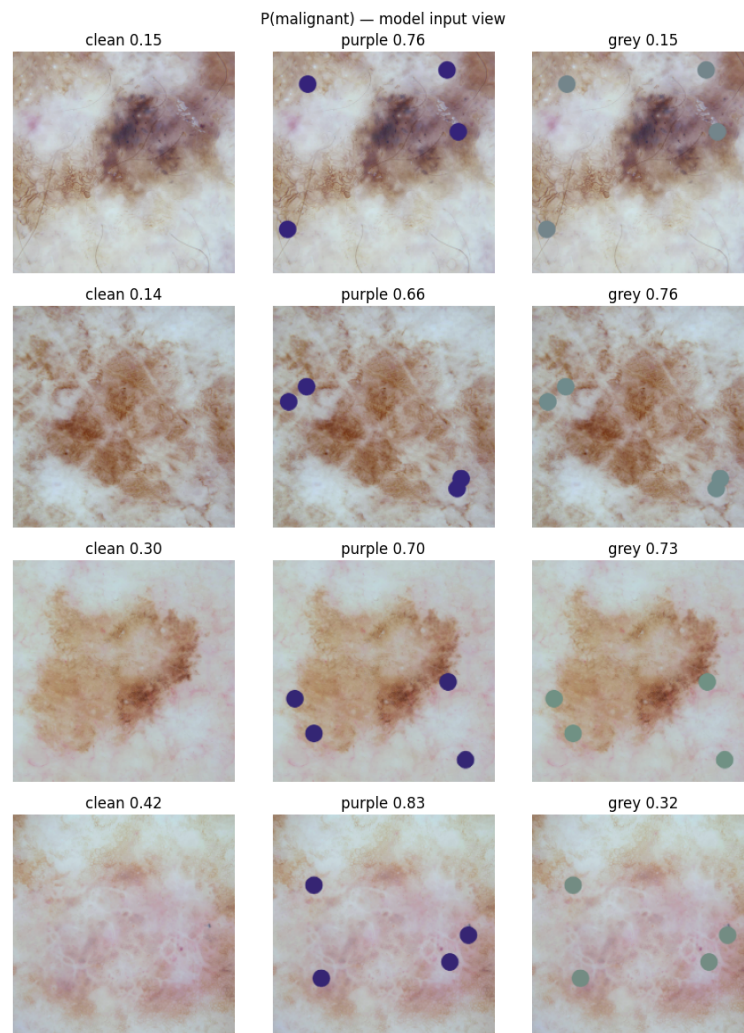


Figure 4. The four images whose $P(\text{malignant})$ rose most under purple ink, shown as the model sees them after colour constancy and cropping. Rows 1 and 4 are purple-specific. Rows 2 and 3 jump just as much with grey dots, so these extreme cases reflect general sensitivity to artifacts. The ink-specific effect is in the aggregate: a small, very consistent push across most images rather than a few dramatic flips.

What I conclude, and what I don't

The model has a **small but consistent ink-specific bias**. The average shift is about one percentage point, so accuracy is essentially unaffected, but at the operating threshold it matters: purple ink sends 7.1% of correctly classified benign lesions into the malignant bucket versus 3.1% for neutral dots. It concentrates in keratoses rather than nevi. None of this is visible in accuracy, macro F1 or calibration error.

Limitations

- Synthetic dots are not real ink. They rule a colour-and-shape cue in, but real marking pens vary in hue, shape and opacity.
- HAM10000 is dermoscopy and mostly lighter skin, and has no Fitzpatrick labels, so I could not test the skin-type interaction that is central to the VCR paper.
- Only Layer 2 (fine-tuned head) was tested. Whether ink also changes the gate's routing decisions is untested.
- One dot placement per image. With $n = 1,064$ the Wilcoxon test detects very small effects, so I report effect sizes alongside p-values.

Next steps I would like to run

- **Skin-type stratified audit on DDI** (Fitzpatrick-labelled), to test whether the ink effect differs by skin tone the way the paper found for LMMs.
- **Real ink**: locate marked images by colour segmentation, inpaint the marks out, and measure the drop.
- **Concept-level analysis**: adapt concept-activation-vector ranking to the ViT's features to look for shortcuts I did not think to test, such as body site or rulers.
- **Mitigation**: ink-dot augmentation during fine-tuning, then re-run the same audit to check the threshold-crossing rate falls.

Licence note: PanDerm weights are CC-BY-NC-ND 4.0; this work is non-commercial research. Full build notes: joan-portfolio-five.vercel.app/builds/amsdds